BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BAHAWALPUR

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

BIOLOGY

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

Paper II (Objective Type) — Inter-A-2021

حیاتیات (معروضی)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** [illegible in source scan]

Note: Four possible choices A, B, C, D to each question are given. Which choice is correct fill that circle in front of that Question No. Use Marker or Pen to fill the circles. Cutting or filling two or more circles will result in Zero Mark in that Question.

Q1. Turgor Pressure is generated by high osmotic pressure of the cell:

(A) Cytoplasm (B) Protoplasm

(C) Vacuole (D) Nucleus

Q2. The active absorption of Sodium in the ascending limb of Henle is promoted by:

(A) ADH (B) ATCH

(C) Vesopression (D) Aldosterone

Q3. Non Surgical removal of Kidney Stone is called:

(A) Dialysis (B) Uremia

(C) Lithotripsy (D) Kidney Transplant

Q4. In Pelvic Region Fusion of four Posterior Vertebrae forms:

(A) Coccyx (B) Sacrum

(C) Pubis (D) Ischium

Q5. Cleavage in fertilized egg results in the formation of:

(A) Gastrula (B) Blastula

(C) Morula (D) Neurula

Q6. Estrogen produced by Ovary inhibits the secretion of:

(A) FSH
(B) LH
(C) ADH
(D) ATCH

Q7. Leaf Abscission is promoted by:

(A) Auxins
(B) Gibberellins
(C) Cytokinins
(D) Abscisic Acid

Q8. The Hormone which releases the lateral buds from apical dominance is:

(A) Auxins
(B) Gibberellins
(C) Cytokinins
(D) Abscisic Acid

Q9. The syndrome in which male has enlarged breasts, obesity and small testes with no sperms is:

(A) Down's Syndrome
(B) Turner's Syndrome
(C) Klinefelter's Syndrome
(D) Jacob's Syndrome

Q10. During Prophase I of Meiosis, Tetrads are formed in:

(A) Leptotene
(B) Zygotene
(C) Pachytene
(D) Diplotene

Q11. All are Stop Codons except:

(A) UAA
(B) AUG
(C) UAG
(D) UGA

Q12. Maternal Foetal Incompatibility can result due to marriage between:

(A) Rh+ male and Rh- female
(B) Rh- male and Rh+ female
(C) Rh+ male and Rh+ female
(D) Rh- male and Rh- female

Q13. Relationship between Shark and Remora attached to it is an example of:

(A) Symbiosis
(B) Mutualism
(C) Parasitism
(D) Commensalism

Q14. All are causes of Green House effect except:

(A) Deforestation
(B) Industrialization
(C) Over Urbanization
(D) Reforestation

Q15. Bacteria can take up recombinant plasmid if they are treated with:

(A) Calcium Chloride
(B) Sodium Chloride
(C) Ammonium Chloride
(D) Barium Chloride

Q16. The Ecosystem in which Soil is Grayish brown, very fertile and rich in organic matter is:

(A) Coniferous Forest
(B) Grassland
(C) Temperate Deciduous Forest
(D) Tundra

Q17. According to Endosymbiont Hypothesis, ingestion of Prokaryotes similar to Cyanobacteria could have developed into:

(A) Mitochondria
(B) Chloroplasts
(C) Nucleus
(D) Dictyosomes

Biology (Subjective) — Inter-A-2021

حیاتیات (انشائیہ)

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Note: It is compulsory to attempt any (8-8) Parts each from Q.No. 2 and Q.No.3 and attempt any (6) Parts from Q.No.4. Attempt any (3) Questions from Part-II. Write same Question No. and its Part No. as given in the Question Paper. Make Diagram where necessary.

Note: Part - I — 22 x 2 = 44

QQ.No.2.

(i) What is Lithotripsy? Give the Mechanism.

(ii) What are Poikilotherms? Give one example as well.

(iii) Define Homeostasis. Give components of Homeostatic Control System.

(iv) Differentiate between Tendons and Ligaments.

(v) Briefly describe the Hematoma Formation.

(vi) Give the composition of Thin Filaments in Skeletal Muscles.

(vii) Give the route of sperms from Testis to Outside in man.

(viii) Define Menopause and Ovulation.

(ix) Differentiate between Prairies and Savanna.

(x) What is Taiga? What conditions do animals face residing there?

(xi) Give the effects of Ozone Layer Depletion.

(xii) Define Eutrophication. How man has speeded up this process?

QQ.No.3.

(i) Describe Feedback Mechanism with an example.

(ii) Define Parasympathetic Nervous System.

(iii) Write a note on Parathyroids Glands.

(iv) What is Complete Dominance?

(v) Define Over Dominance.

(vi) Differentiate between Homozygous and Homozygote.

(vii) Write a note on Recombinant DNA Technology.

(viii) How can we get a Gene of Interest?

(ix) How Plant Health can be promoted by Transgenic Bacteria?

(x) What is a Niche?

(xi) Define Food Web.

(xii) Define Lichens in detail.

QQ.No.4.

(i) What are Intercalary Meristems?

(ii) What is Discoidal Cleavage?

(iii) Define Point Mutation.

(iv) Compare Euchromatin with Heterochromatin.

(v) What are Okazaki Fragments?

(vi) What is Metastasis?

(vii) What changes occur in Cell during anaphase of Mitosis?

(viii) Define the term Homology. Give example.

(ix) What is the role of Migration in affecting Gene Frequency?

> Note: Part - II

QQ.No.5. (Marks 4) (a) Discuss Counter-Current Multiplier with reference to concentration of excretory products.

Q. (Marks 4) (b) Write a note on the Nitrogen Cycle.

QQ.No.6. (Marks 4) (a) Describe the types of Joints on the basis of Structure.

Q. (Marks 4) (b) How Cells use RNA to make Proteins?

QQ.No.7. (Marks 4) (a) What is Synapse? Diagrammatically mention how nerve impulse passes through a synapse?

Q. (Marks 4) (b) Describe importance of Forests for human life.

QQ.No.8. (Marks 4) (a) Describe the Female Reproductive Cycle in Humans.

Q. (Marks 4) (b) Define and explain Incomplete Dominance with example.

QQ.No.9. (Marks 4) (a) Describe Growth Correlation in detail.

Q. (Marks 4) (b) Discuss Theory of Natural Selection and Adaptation.